

Ji Bing Ni

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EDUCATION

Purdue University

Bachelor of Science in Computer Science

West Lafayette, IN

Aug. 2022 - May 2026

EXPERIENCE

Software Developer

Jan. 2026 – Present

Purdue Vertically Integrated Projects

West Lafayette, IN

- Enabled **secure MCP server access** by implementing **token-based authentication** on **HTTP/SSE calls**
- Extended **geoscience AI assistant** with **account-linked actions** through **web-authenticated MCP sessions**
- Improved **MCP server reliability** by expanding **test coverage** between **Fastify server** and **MCP server**

Software Developer

Jan. 2025 – Aug. 2025

Purdue University – AIM Research Team

West Lafayette, IN

- Achieved real-time audio to sheet music alignment with **OSMD API** and **Dynamic Time Warping**
- Developed a music evaluator app with **React Native** and **TypeScript** for improving musicians' performance
- Reduced potential hosting costs by migrating app from **client-server** to a **serverless architecture**
- Enhanced app scalability by **approx. 10%** through creation of **4+ reusable UI components**
- Improved production readiness by **approx. 15%** via **Expo Go** for real-device testing on **iOS** and **Android**

Systems Programming Teaching Assistant

Jun. 2025 – Present

Purdue University – Department of Computer Science

West Lafayette, IN

- Led **30+ labs** for **40+ students**, improving debugging efficiency and conceptual understanding in **C/C++** via projects on a **custom shell interpreter**, **malloc**, **Bash scripting**, and **HTTP server development**
- Increased project completion rates by **approx. 20%** for **40+ students** through **20+ office hours**
- Improved comprehension for **40+ students** by **approx. 15%** through grading labs & exams with feedback

Undergraduate Data Science Researcher

Aug. 2024 – May 2025

The Data Mine - Purdue University

West Lafayette, IN

- Reduced call center reliance by building an **AI chatbot** for **Wabash** in an **8-member Agile team**
- Enhanced response quality using **prompt engineering** with an **NLP model** and **machine learning**
- Enabled scalable chatbot interactions by implementing a front-end with **RESTful API** and back-end integration
- Strengthened reliability across **20+ devices** through **5+ API tests** validating connectivity and data accuracy

PROJECTS

PokéWorld | *React, Express.js, Node.js, MongoDB, REST APIs*

May 2024 – Present

- Developed an interactive Pokémon platform with a full-stack **MERN app** powered by **PokéAPI (1,000+ Pokémon)**, **Pokémon TCG API (150+ sets, 10k+ cards)**, and **TMDB API (20+ movies)** for fan exploration
- Secured accounts with **bcrypt**-hashed passwords in **Express.js** and **MongoDB**, built to support **1,000+ users**
- Designed a scalable UI with **4+ reusable components** using **Material-UI** and **TailwindCSS**
- Reduced API calls by **approx. 20%** via **debounced search**, **local cache**, and **on-demand pagination**

BoilerFinds | *React, Flask, PostgreSQL, Bootstrap, TMDB API, OpenStreetMap API*

Feb. 2025 – May 2025

- Engineered a full-stack platform using **React**, **Flask**, and **PostgreSQL**, supporting large-scale user discovery
- Enhanced UX through a responsive **Bootstrap** UI, powering **10+ pages** and **20+ reusable components**
- Implemented a scalable **RESTful API** with **Flask** and **PostgreSQL**, supporting data for **1M+ users**
- Enabled **1M+ movie discovery** and interactive restaurant maps via **TMDB** and **OpenStreetMap APIs**

Custom Unix Shell | *C++, Lex, Yacc, POSIX*

Jan 2024 – May 2024

- Built a **Unix-like shell in C++** with **command execution**, **pipes**, **I/O redirection**, and **background jobs**
- Enabled advanced features such as **expansions**, **subshells**, and **wildcards** using a **Lex/Yacc-based parser**
- Strengthened shell reliability with **signal handling** for events such as process termination and user interrupts

TECHNICAL SKILLS

Languages: Java, Python, C, C++, JavaScript, TypeScript, SQL, R, HTML, CSS, Tailwind CSS

Frameworks: ReactJS, React Native, AngularJS, Express.js, Node.js, MongoDB, PostgreSQL, SQLite, Flask, Bootstrap

Developer Tools: Git, GitHub, Docker, Azure DevOps, LaTeX, VS Code, IntelliJ, Eclipse

Libraries: pandas, NumPy, Matplotlib, BeautifulSoup, SQLAlchemy